

Lab Report for Lab Electricity

DC Power Sources

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Lab responsible:

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1 INTRODUCTION

Describe briefly, clearly and yet precisely the goal of the experiment / measurements and the strategy with which you will achieve that goal.

2 BACKGROUND/ THEORY

Here we refer to the lab text and the lecture text. [1] [2]

3 METHOD & MATERIALS

Specifically for this experiment:

Provide a brief description of the measurement setup and the measurements performed.

3.1 Experimental Set-up determination of internal resistance

3.2 Experimental Set-up maximum power transfer

4 DETERMINATION OF INTERNAL RESISTANCE

4.1 Measurement results

First add one or more sentences that already provide the main measurement results as text or refer to a table. Then add the measurement results in a compact and clear table, possibly clarified by appropriate graphs. If legibility is compromised due to the large number of measurement results, certain measurements in the table can also be included in the appendices. A table always has a caption above it and is numbered. Provide an informative, clear caption.

The header of a table consists of at least 2 lines:

- 1st line: the default symbol for the quantity

- 2nd line: the unit used in brackets, possibly preceded by a prefix or an appropriate power of 10. Only SI units are allowed.

Numbers are presented as simply as possible. Make a suitable choice of the unit and use prefixes such as mega or giga. Only significant figures are mentioned.

Avoid spreading a table over two pages. If a table does not fit on one page due to its size, repeat the header of the table on the next page.

Specifically for this experiment:

Make one clear table with all measured voltages (Source 1, Source 2, series, parallel) in function of the supplied current.

Also mention the directly measured values of the additional "internal" resistors of the individual sources connected in series.

4.2 Graphs

Graphs follow the same layout method as figures. Make sure that charts are sufficiently large and legible. Think about the scale and the choice of units. Make sure that there is a scale along each axis: 3 to 10 graduation marks with round numbers at regular intervals (2 to 3 significant figures). Do not include readings along the axes. For each axis, place the symbol of the quantity plotted and in brackets the unit used. Make sure the readings themselves are clearly visible.

If possible, draw a line of best fit or a theoretical line through the measurement points (e.g. via a regression analysis or least squares method). Include the corresponding equation (and R^2) in the figure. Provide a legend with meaningful names if there are several curves on one graph. Centrally align the figures. And of course, provide an informative caption with a number which is referred to in the text.

Specifically for this experiment:

Graphs of voltage in function of current: Create four different graphs with the measured voltage as a function of the current supplied. Add the trend line through the measuring points.

4.3 Calculations

Present the calculations of the measurements in an orderly manner. If several similar calculations are performed, give at least 1 complete, clearly worked out example calculation and an error calculation for all calculated quantities. Clearly indicate for which measurement value from the table the calculations are made. First work out the calculations using variables/symbols, only then enter the numbers. If necessary, provide explanations (for example, textually explain the mathematical technique used). Always put the final result in standard notation.

Specifically for this experiment:

Use the equation of the trendline to determine the V_{ems} , R_i and I_{sc} for all four sources. Summarize the results in a separate table. Pay attention to the number of significant digits!

For source 1, also calculate the error on all these variables using the statistical error on the trendline parameters, as described in the introduction to the dynamics lab. Also calculate the error on the directly measured value of the V_{ems} and the additional "internal" resistance of that source that is connected in series.

Calculate the theoretically predicted values of V_{ems} , R_i and I_{sc} for the series and parallel circuit of the sources, using the graphically determined values of both sources separately.

4.4 Discussion

In this section, repeat the central questions / goals from the lab assignment and formulate an answer, possibly grouped per sub-experiment. Interpret your results and decide: has the purpose of the experiment been achieved? Try to explain any discrepancies. For example, discuss the following:

- Accuracy and precision of the measuring method (s): Which method affects the measurement error of the final result most significantly? What causes the biggest uncertainty?
- Correctness of the final result: Avoid terms such as "reasonably good" or "about"; either a measurement is correct with the expected value within the measurement error, or not.
- Give a final assessment of the obtained results in comparison with the results of others, for example from previous experiments, other students or previous literature.

If your research concerns a relationship between 2 or more physical quantities, describe the shape and strength of the relationship found and, as far as possible, explain the comparison with the theoretically expected relationship. Point out the slope of the line and / or its physical interpretation. Beware of terms such as "directly proportional", "inversely proportional", "exponentially increasing/ decreasing". These have one specific physical meaning; use it correctly!

Suggestions: If necessary, formulate tips for improving the measurement methods and / or alternative measurement setups. Be sure to give suggestions for better measuring methods or accuracy of the calculations if the test did not go as expected.

Specifically for this experiment:

For each of the single sources, compare the open circuit voltage determined from the trendline to the directly measured value, and the internal resistance determined from the trendline to the resistor connected in series with the battery.

Is it possible to determine the real internal resistance of the battery itself from these measurements? Explain!

For the series and parallel connection of the sources, compare the graphically determined values with the theoretical predictions.

Draw relevant conclusions from the measurements and calculations you have made.

5 MAXIMUM POWER TRANSFER

5.1 Measurement results

Specifically for this experiment:

Arrange the measurement results in a well-organized table (nominal values of R_L , measured values of R_L , voltage V_L and power P_L).

5.2 Graphs

Specifically for this experiment:

Create two graphs in which you show the power in function of the load resistance.

5.3 Calculations

Specifically for this experiment:

For one of the sources, calculate the error on the power obtained for the measuring point where the power is at its maximum.

5.4 Discussion

Specifically for this experiment:

Discuss the curves obtained and formulate relevant conclusions from the measurements and calculations you have made.

6 CONCLUSION

Briefly repeat the most important finding and / or final assessment of the results obtained in comparison with results from previous experiments, from others or from literature. Do not put new information here.

7 BIBLIOGRAPHY

Put the list of sources here. Use a reference manager. The default reference style is IEEE or APA. However, this can be deviated from for the various assignments if the teacher mentions this explicitly. So always check this carefully.

[1]

[2] C.K. Alexander en M.N.O. Sadiku, *Fundamentals of Electric Circuits*, International edition. NY, USA: Mc-Graw Hill Education: 2017